Mini-Project (ML for Time Series) - MVA 2025/2026

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1 Introduction and contributions

Audio Source Separation is a long-standing problem in signal processing, often illustrated by the "cocktail party problem," where a listener attempts to focus on one speaker among several. Traditional methods such as Independent Component Analysis (ICA) and Non-Negative Matrix Factorization (NMF) have been widely used but remain limited when dealing with complex, real-world mixtures. Deep generative models, notably Variational Autoencoders (VAEs), now offer a principled probabilistic framework for learning latent representations of audio signals. This work fits within this modern line of research by examining how different latent-space regularization strategies; classical VAE regularization via the KL Divergence, the extended constraint of the β-VAE, and the non-parametric divergences used in the ITL-AE affect the model's ability to capture meaningful audio structure and improve source separation performance.

The task addressed is the separation of two audio sources, where one source (S_1) is treated as the target signal and the other (S_2) as interference. Given a mixture $M = S_1 + S_2$, the model receives the spectrogram of the mixture (or, in alternative experiments, clean data) as input and is trained to reconstruct the clean target spectrogram. This setup, applied across VAE, β-VAE, and ITL-AE architectures, corresponds to a denoising configuration and can be naturally extended to mixtures involving more than two sources.

With regard to the division of labour, the two students first read the article [1] independently. Matéo then implemented a standard Variational Autoencoder, followed by an improved version incorporating RNNs. Two training strategies were tested: one in which the models were trained on clean separated sources and later applied to noisy mixtures, and another supervised approach where the models were trained directly on noisy sources. Maxence implemented the same models for two VAE variants presented in this report: the β-VAE and the ITL-AE. For the report, both students wrote approximately two sections each, with a larger writing workload for Matéo; to compensate, Maxence prepared the presentation slides. No source code matching the article's implementation was found, so all code was written from scratch. The original article applied VAEs to source separation but remained vague and imprecise, providing no hyperparameters, no description of preprocessing, and no clarification of the exact task (reconstructing from noisy signals or from clean ones for example). This required testing several scenarios and searching for suitable hyperparameters. Improvements on the original method included post-processing and a grid search for optimal parameters.

2 Method

Let x denote an observed random variable, representing here a specific time-slice of a spectrogram. Let $p_\theta(x)$ be its probability density function, where θ represents the model parameters.

We assume that the distribution of the vector x is defined as the marginal of a joint distribution involving a lower-dimensional variable z called the latent vector. Therefore, it is expressed as $p_\theta(x) = \int p_\theta(x|z)p(z)dz$.

The Variational Autoencoder (VAE) has two objectives. First, it aims to estimate the latent vector z corresponding to a given observation x. This involves estimating the distribution $q_\phi(z|x)$ using the encoder. Second, it must be able to reconstruct an observation x from a latent vector z, which implies estimating the distribution $p_\theta(x|z)$.

It can be shown that for any distribution $q_\phi(z|x)$, the inequality

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log(p_\theta(x|z))] - \mathbb{D}_{KL}(q_\phi(z|x)||p(z)) \leq \log(p_\theta(x))$$

holds. Here, $\mathbb{D}_{KL}(p||q)$ denotes the Kullback-Leibler (KL) divergence between p and q, defined as $\mathbb{D}_{KL}(p, q) = \int_{-\infty}^{\infty} p(x) \log(p(x)/q(x))\, dx$ Thus, maximizing the Evidence Lower Bound (ELBO), denoted as $\mathcal{L}$, with respect to θ and ϕ increases the model's likelihood.

The first term of L corresponds to a negative reconstruction loss; it is maximized when the data reconstructed by the VAE matches the input data. In practice, the distribution $p_\theta(x|z)$ is modeled by the VAE decoder. Consequently, this term is substituted with a Mean Squared Error (MSE) loss calculated between the encoder input vector x and the decoder output vector $\hat{x}$.

The second term of L, given by the KL divergence between p and q, represents a distance between the two distributions. This term is maximized when p closely resembles q. Maximization of $\mathcal{L}$ is performed using the reparameterization trick to address the expectations.

The initial version of the Variational Autoencoder presented in the reference article approximates $\mu_\phi(x)$ and $\sigma_\phi(x)$ using Multi-Layer Perceptrons (MLPs). However, as our work involves time-series data, the vectors x and z are time-dependent. In order to account for longer-term temporal dependencies between those vectors, we decided implement a variant of the VAE known as the Variational Recurrent Autoencoder (VRAE) [2], which uses recurrent neural networks to capture long-term temporal dependencies.

In addition, the article studied proposes several VAE variants. We implemented the two that were mentioned the most. The first one, called the β-VAE [3], simply introduces a regularization parameter into the ELBO. Thus, the ELBO becomes:

$$\mathbb{E}_{q_\phi(z|x)}[\log(p_\theta(x|z))] - \beta\, \mathbb{D}_{KL}(q_\phi(z|x) \parallel p(z)).$$

When $\beta = 1$, it is the same as the standard VAE. When $\beta > 1$, it applies a stronger constraint on the latent bottleneck and limits the representation capacity of z. For some conditionally independent generative factors, keeping them disentangled leads to the most efficient representation. Therefore, a higher β encourages more efficient latent encoding and further promotes disentanglement. At the same time, a higher β may introduce a trade-off between reconstruction quality and the degree of disentanglement.

The second variant, called the "Information-Theoretic Learning Autoencoder," (ITL AE) [4] presents two main differences. One is that it does not use the reparameterization trick. The second is that, instead of using the KL Divergence, it uses other Divergence measures such as the Cauchy–Schwarz Divergence and the Euclidean Divergence, which are estimated through kernel density estimation (KDE). Furthermore, the following quantity would be equivalent to ELBO:

$$L(x, \tilde{x}) - \lambda R(\text{Enc}, p),$$

where L is the reconstruction loss as before, R is the Euclidean or Cauchy–Schwarz Divergence, p is the chosen prior, Enc is the encoder, and λ controls the magnitude of the regularization.

To apply VAEs (or all other variants) to the task of source separation, the reference article indicates that a dedicated VAE must be employed for each distinct source. These models operate exclusively on the magnitude spectrograms of the audio signals, while the phase information is kept constant. We investigated two distinct methodologies for implementing VAEs in this context.

The first method is a generative source modeling method. It involves training a separate VAE for each source to reconstruct signals belonging strictly to that specific source domain (without seeing other sources or mixtures). The underlying hypothesis is that the VAE learns to encode the spectral characteristics intrinsic to its training source. During the source separation phase, we aim to identify the specific latent vectors whose corresponding VAE outputs sum up to approximate the observed mixture. Consequently, source separation becomes an iterative optimization problem performed at inference time.

The second method is a supervised learning approach. It involves training each VAE to reconstruct the clean source directly from input mixtures. This approach requires modifying the reconstruction loss so that it compares the model output to the underlying ground-truth source and not the input mixture. This method is the one used in the reference article.

To reduce the noise present in the reconstructed spectrograms, the outputs of the VAEs are used to apply a Wiener filter to the mixtures to reconstruct the separated signals. The estimated spectrogram $\hat{s}_i$ of source i is thus given by

$$\hat{s}_i = \frac{s_i^1}{s_1^1 + s_2^1} \odot s_{mixt}$$

where s_{mixt} is the spectrogram of the input mixture, s_i^1 is the spectrogram reconstructed by a VAE for source i and $\odot$ is the element wise multiplication.

3 Data

We selected data from two distinct datasets, each representing a specific source type. The first is LibriSpeech [5], a dataset comprising recordings of individuals (both male and female) reading newspaper articles. This dataset constitutes Source 1, that we consider as our target signal. The second dataset, serving as Source 2, is the wave2vec2-vd-brid-sound-classification-dataset [6], which contains recordings of birds in highly noisy natural environments. This source is considered as a noise. Both sources were recorded at a sampling rate of 16 kHz. To accelerate model training, we restricted the dataset to 400 signals, split into a 80% training set and a 20% testing set.

The original data exhibited variable durations. However, model training requires conversion into tensors of fixed dimensions. Consequently, we decided to standardize the time sequences to a duration of 4 seconds. Input sequences exceeding this limit were truncated, while shorter sequences required padding. A zero-padding was applied randomly at the beginning and at the end of the sequences until the 4-second duration was reached.

To facilitate training, the data was normalized by the maximum value, scaling the peak amplitude to 1. While this preprocessing method, which was suggested in the reference article, facilitates VAE training, it relies on the assumption that the data is free of outliers. In the presence of outliers

with significant amplitude peaks, this method would compress the signal of interest toward zero, and therefore an outlier detection and removing step would be required.

Source mixtures were constructed by combining signals from Sources 1 and 2 at a fixed Peak Signal-to-Noise Ratio (PSNR) of 10 dB, with Source 2 treated as a noise. For each pair, the power of both signals was calculated to rescale the signals appropriately before summing them. This is a simplifications, as in real mixtures, the PSNR is not known or constant in general.

Finally, the audio signals were transformed into spectrograms using the Short-Time Fourier Transform (STFT). The STFT window size was set to 512 points with a hop length of 128 points, resulting in an overlap of 384 points to ensure higher resolution of the spectrograms. The phase and magnitude components of the STFT were separated. The magnitude only is used by the VAEs for source separation. This magnitude was converted to a logarithmic scale using the log1p function, as a logarithmic scale aligns more closely with human auditory perception.

Since VAEs generate synthetic data, a post-processing step is required for the model outputs. Given that the VAEs are tasked with producing spectrograms (magnitude matrices), the output values must theoretically be non-negative. However, depending on the specific network architecture employed, the model may occasionally generate negative values. In such instances, these invalid entries are treated as missing values and are strictly replaced with zeros.

The spectrograms generated by the VAEs subsequently serve as oracles for a Wiener filter, enabling the estimation of the target signals as detailed in the Methods section. To reconstruct the final audio, the magnitude spectrograms obtained from the filter are combined with the phase information from the original mixture. Finally, an Inverse Short-Time Fourier Transform (iSTFT) is applied to recover the time-domain audio signal.

4 Results

In accordance with the studied article, Table 1 shows the performance of our models, which are evaluated in terms of SDR (Signal to Distortion Ratio), SAR (Signal to Artifact Ratio), and SIR (Signal to Interference Ratio). These are common measures to evaluate speech source separation. The estimated signal can be written as follows:

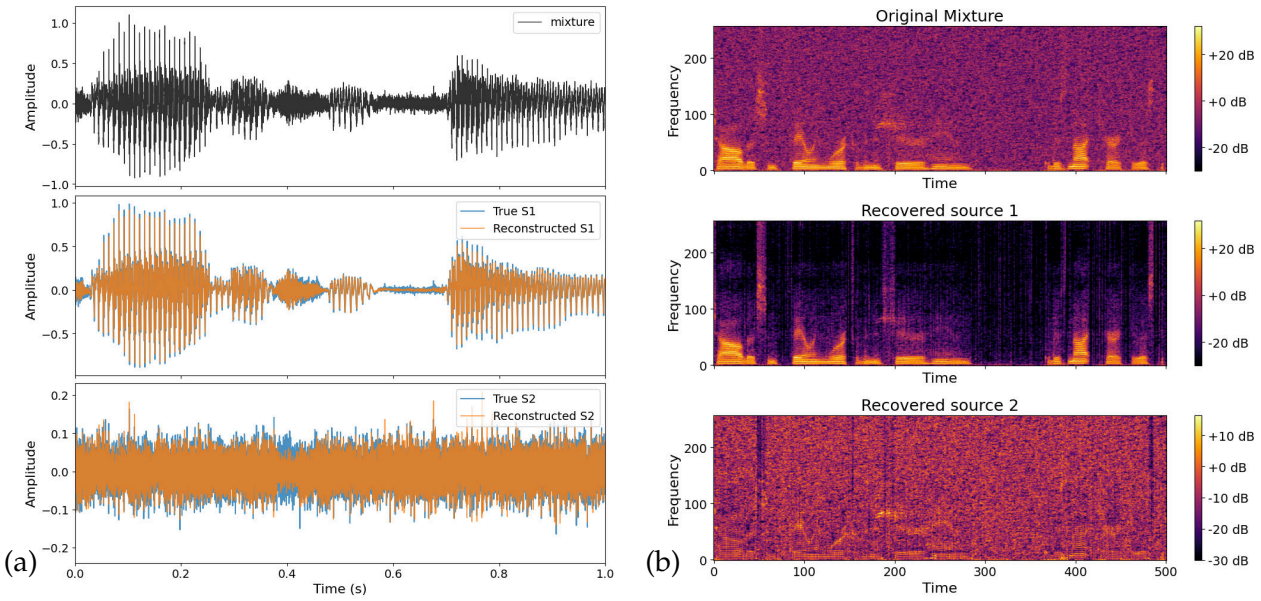
$$\hat{s}(t) = s_{\text{target}}(t) + e_{\text{interf}}(t) + e_{\text{noise}}(t) + e_{\text{artif}}(t)$$

where $s_{\text{target}}(t)$ is a deformed version of $s(t)$, $e_{\text{artif}}(t)$ is an artifact term, $e_{\text{interf}}(t)$ denotes the deformation of the signals due to interference from unwanted signals, and $e_{\text{noise}}(t)$ is a deformation of the perturbing noise. Thus:

$$\text{SDR} = 10\log_{10}\frac{\|s_{\text{dist}}\|^2}{\|e_{\text{interf}} + e_{\text{noise}} + e_{\text{artif}}\|^2}, \quad \text{SIR} = 10\log_{10}\frac{\|s_{\text{dist}}\|^2}{\|e_{\text{interf}}\|^2}$$
$$\text{SAR} = 10\log_{10}\frac{\|s_{\text{dist}} + e_{\text{interf}} + e_{\text{noise}}\|^2}{\|e_{\text{artif}}\|^2}$$

For the reconstruction of Source 1, which corresponds to the speech signal, we can observe that(cf. Table 1) in general, all three metrics are higher when the models are trained on mixed signals

rather than on isolated sources. However, complexifying the autoencoder architecture by introducing RNNs does not particularly improve the results. We can also note that the β-VAE variant slightly outperforms the basic VAE version when the models are trained on mixed signals (without changing any parameters between the two, only adapting the loss function). This is consistent with the results from the article, which leads us to believe that this is indeed the framework in which they trained their models, as this was not clearly specified. Finally, in the article, the authors claim that the ITL_AE version has worse results in terms of SAR and better results in terms of SIR [1, p. 46]. This is verified in our case for SAR but not always for SIR. We believe that the hyperparameters of our models for ITL_AE are not optimal, but the computational resource limitations prevented us from doing better. In particular, the calculations of Euclidean and Cauchy-Schwarz divergences were very costly during training.



Field: a mixed signal of test set - VAE

Figure 1: Spectrograms and waveforms of mixed signal and separated sources (speech and noise)

The reconstructed waveforms and spectrograms (Figure 2) show that both sources are reasonably well reconstructed. The temporal structure and spectral characteristics are preserved. However, we observe that the amplitude of the speech signal is sometimes overestimated, resulting in slightly higher energy than the original source.

Listening to the reconstructed audio confirms the visual observations. The speech is intelligible and the main characteristics are preserved. However, some residual noise remains in the separated speech signal, indicating incomplete source separation. This is consistent with the SIR values obtained, which suggest that some interference from the noise source persists in the speech reconstruction. Our results closely match those reported in the original article when using models trained on mixed signals without RNNs. This validates that the model architecture and underlying assumptions are well-suited to our data and the blind source separation problem. We also were able to verify that deep learning methods outperform the classical ICA method as claimed by the authors [1, p. 2]. Unfortunately we lacked time to implement the NMF methods for comparison. It should be noted that the authors themselves did not provide such comparisons in their article, nor did they include qualitative descriptions of the audio quality obtained by their models.

References

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A Technical details about our models

To simplify calculations, we assume a Gaussian prior for z, such that $p(z) = \mathcal{N}(z; 0, I)$, and that the posterior approximation is given by $q_\phi(z|x) = N(z; \mu_\phi(x), diag(\sigma^2_\phi(x)))$. Computing the gradient of L is challenging because the sampling operation (represented by the expectation over $q_\phi(z|x)$) is non-differentiable. To address this, we employ the reparameterization trick by setting

$$z = \mu_\phi(x) + diag(\sigma_\phi(x))\varepsilon$$

where $\varepsilon \sim \mathcal{N}(0, I)$. This reparameterization trick allows backpropagation through the sampling of z step that is required to approximate the expectation.

In the initial version of the Variational Autoencoder, as presented in the reference article, the functions $\mu_\phi(x)$ and $\sigma_\phi(x)$ are approximated by Multi-Layer Perceptrons (MLPs) within the VAE encoder. In our specific case, we selected ReLU activation functions and three linear layers with an intermediate dimension size of 128. The decoder features an architecture symmetric to the encoder, comprising a three-layer MLP with an intermediate dimension of 128 and ReLU activations. In our VRAE variant of the VAE, we captured long-term temporal dependencies by integrating recurrent networks into both the encoder and decoder. We chose to implement this variant using single-layer Bidirectional LSTM recurrent networks in order to check if adding this longer term temporal dependency would improve the results. Due to computational time limits, we decided to train all our models on 200 epochs.

B Additional Visualizations of the results

Models	Description	SDR(dB)	SIR(dB)	SAR(dB)
VAE	Models trained on separate signals (unsupervised)	3.20	16.58	3.83
β-VAE		0.93	15.18	2.01
ITL_AE		-19.26	2.56	-15.81
VAE	Models trained on mixed signals (supervised)	6.88	11.13	10.08
β-VAE		11.08	18.67	12.55
ITL_AE		6.88	11.13	10.08
VRAE	RNN-based models trained on mixed signals (supervised)	3.38	12.63	4.53
β-VRAE		7.80	20.70	8.29
ITL_RAE		5.09	15.93	5.94
ICA	Méthode classique baseline	2.21	15.14	2.67

Field: Test set - Speech signals (Source 1)

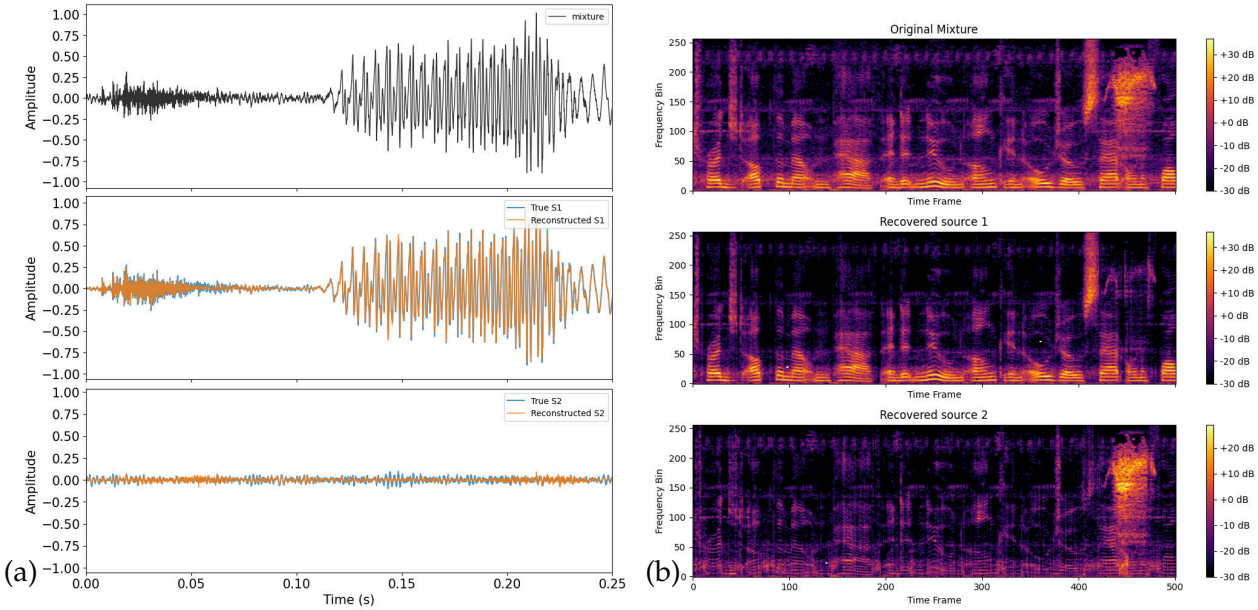
Table 1: Source separation performance metrics for speech signal reconstruction (Source 1)

Models	Description	SDR(dB)	SIR(dB)	SAR(dB)
VAE	Models trained on separate signals (unsupervised)	-8.45	2.63	-5.57
β-VAE		-10.40	2.88	-7.47
ITL_AE		-21.37	-1.73	-16.09
VAE	Models trained on mixed signals (supervised)	-5.43	-4.48	9.87
β-VAE		3.05	8.78	5.56
ITL_AE		-5.43	-4.48	9.87
VRAE	RNN-based models trained on mixed signals (supervised)	-7.34	-5.34	4.90
β-VRAE		5.09	15.93	5.94
ITL_RAE		6.27	17.71	6.95
ICA	Méthode classique baseline	-5.38	-1.77	1.50

Field: Test set - Speech signals (Source 2)

Table 2: Source separation performance metrics for noise signal reconstruction (Source 2)

In Table 1 and Table 2, the higher the metrics, the better the model reconstructs the signal.



Field: a mixed signal of test set - β-VAE

Figure 2: Spectrograms and waveforms of mixed signal and separated sources (speech and noise)